Automatic Alignment of Handwritten Images and Transcripts for Training Handwritten Text Recognition Systems

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Automatic Alignment of Handwritten Images and Transcripts for Training Handwritten Text Recognition systems

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***Abstract*—State-of-the-art Handwritten Text Recognition techniques are based on statistical models such as hidden Markov models or recurrent neural networks for optical modeling of characters and N-grams for language modeling. These models are trained using well known, learning techniques: Expectation-Maximization, backpropagation, etc. Therefore, training data is needed to build these models. In the case of the optical models the training data consist of text line images with their corresponding transcripts. When the transcript of a handwritten document is available, putting in correspondence automatically the physical lines in the images with the lines of the transcripts is not an easy task. We present a method for automatically aligning handwritten text images and their respective transcripts. The approach automatically segments the images into lines and then recognizes them. An alignment confidence is obtained using the Levenshtein distance between the recognition results and the transcripts. The most confident lines are then used for training. Experiments carried out using a historical document present encouraging results.**

***Keywords*-Automatic Alignment, Text line segmentation, Handwritten text recognition,**

I. Introduction

Huge amount of historical handwritten documents are being made available world wide. To make these documents really useful, they need to be transcribed by means of Handwritten Text Recognition (HTR) systems.

Traditional OCR techniques are useless, since characters can not be isolated automatically in these images. Therefore, holistic, segmentation-free Handwritten Text Recognition (HTR) techniques are needed [1]. State-of-the-art segmentation-free HTR techniques run at line level and are based on statistical models such as hidden Markov models [1] or recurrent neural networks [2] for optical modeling of characters and N-grams for language modelling [3].

These models are trained using well known, powerful learning techniques like the Expectation-Maximization or backpropagation algorithms. Therefore, training data is needed to build these models. In the case of the optical models the training data consist in text line images with their corresponding transcripts. When the transcripts are not available, then they are produced manually or by using assisted transcription systems [4]. However, there is a large quantity of historical documents for which transcripts are available. These transcripts could be used to train optical models if they are adequately aligned with the lines in the corresponding page images. Thus HTR systems could greatly benefit if we could automatically add new training corpus by solving the problem of the correspondence of the physical lines in the images with the lines in the already made transcripts.

Clearly, only when the number of transcripts are large enough, it becomes interesting to try to use them for training. This requisite makes it not feasible to perform the required alignments manually, and therefore automatic procedures are needed. However, this is not an easy task. First, it is necessary to annotate the geometric information of the main layout parts such as text blocks and lines. Then, assuming the transcripts are given at line level [5] (i.e., with synchronized line breaks, which is often the case), the problem is how to create the correspondence between the lines in the transcripts and the lines in the images. But, even in this case, different critical problems arise: first, just an additional line in the transcripts or in the images can produce a shift in the remaining lines. Second, even if the number of transcribed lines and line images are the same, the pairing may depend on a non uniform reading order.

In the case of a perfect line image segmentation and if some "natural" reading order can be assumed in the pages, then force alignment solutions can be used to find the correct alignment. This approach has been extensively studied in the literature [6], [7], [8], [5], [9]. In [9] they proposed an alignment based on a HMM recognition system. A sequence of analytical features are extracted from the text line images of a page and they are matched with a page HMM created according as per the given transcription, including spelling variants and modeling deletions. In [10] an alignment system for inaccurate transcription based on HMMs is presented. First, they perform the recognition of the text line images using a lexicon containing only the words of the page. Then a string edit distance is used as the cost function to align the recognition output with the transcriptions. Finally, keyword spotting is used to recovered lost words.

All the above described systems assume that the line positions in a page are known or can be reliable retrieved. However, these methods are useless for the problem tackled in this paper. On the one hand the automatic line detection systems are not error-free [11], resulting in pages with undetected lines or lines detected in incorrect positions. On the other hand, the available transcriptions for historical documents sometimes do not match the image

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content exactly, the main causes are missing line breaks, missing transcriptions of small pieces of text like catchwords or page numbers, and non diplomatic transcriptions (expanded abbreviations, corrected mistakes, ...).

In this paper we present a method for automatically aligning handwritten text images and their respective transcripts with synchronized line breaks. The approach automatically segments the images into blocks and lines and then makes an alignment. To carry out the alignment, each line of a page is automatically recognized using a special language model created only with the transcripts of the corresponding page. Then, an alignment confidence is computed based on the Levenshtein distance between the system result and the transcript. These Image-line / transcript-line pairs with high alignment confidence score can be used for improving HTR models by training them with more data, and so evaluate the impact in the final HTR performance. A similar work has been recently studied with promising results in [12].

In Section II the approach used for robust text line segmentation is explained. Then, Section III explains how the alignment is carried out. Section IV gives a brief overview of the complete HTR system considered here. The experimental framework and the results are described in Sections V and V-D. Finally, some conclusions are drawn in Section VI.

II. Layout Analysis

Page layout analysis is performed in two steps: text block detection, followed by text line detection and segmentation.

To start with, each page image is preprocessed in order to reduce the noise and to remove the variance in the background [13]. Then, the block and lines are detected as described next.

A. Text Block Detection

Text blocks in each page are located though a simple procedure based on horizontal and vertical line detection using of enhanced profiles. An example of a detected text block can be seen in Fig. 1.

Then, the block information obtained is used to eliminate all issues outside the text blocks such as speckle, borders, margin issues, etc.

B. Text Line Detection and Segmentation

Here we follow the statistical text line analysis and detection (TLAD) approach presented in [13]. The method is based on hidden Markov Models (HMMs) and finite state or N-gram vertical layout models.

Each page image is represented as a sequence of feature vectors which conveys information about the vertical page structure. These features consist of horizontal projection profiles computed in several vertical slabs of the page [13].

The TLAD problem is formulated as the problem of finding the most likely sequence of line class label hypotheses, $\hat{\mathbf{h}} = \hat{h}_1, \hat{h}_2, \ldots, \hat{h}_n$, for a given handwritten page image (or image block), represented as a sequence

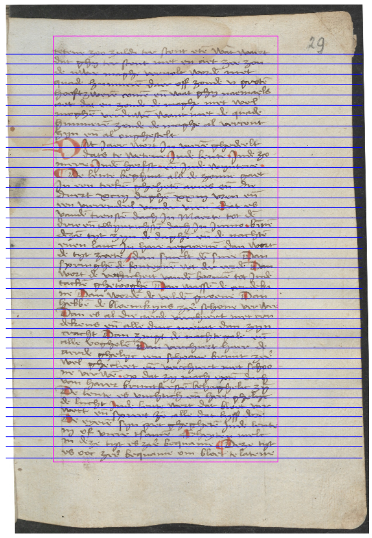

Figure 1. Example of text block and text line detection results.

of "observations" (i.e., feature vectors) $\mathbf{o} \equiv \vec{o}_1^{\,L} = \vec{o}_1, \vec{o}_2, \ldots, \vec{o}_L$; that is:

$$\hat{\mathbf{h}} = \arg\max_{\mathbf{h}} P(\mathbf{h} \mid \mathbf{o}) = \arg\max_{\mathbf{h}} P(\mathbf{h})\, P(\mathbf{o} \mid \mathbf{h}) \quad (1)$$

where $P(\mathbf{o} \mid \mathbf{h})$ is a *line vertical shape model* and $P(\mathbf{h})$ is a *vertical layout model* (VLM). $P(\mathbf{o} \mid \mathbf{h})$ is approximated by HMMs, while $P(\mathbf{h})$ is modelled by an N-gram, or by a more explicit, finite-state model representing a-priory restrictions of how the different classes of line regions can be concatenated to form a valid text page. Following ideas presented in [13], only three horizontal line region labels are considered; namely Blank Space (BS), Normal (base)Line (NL) and Inter Line (IL). Accordingly, a very simple finite-state VLM is used which allows for any concatenation of pairs NL-IL, surrounded by BS regions.

In TLAD, we are interested not only in adequately labelling each horizontal region, but also in actually determining their corresponding *vertical position* inside the page. As discussed in [13], Eq. (1) can be approximately rewritten so as to obtain both the best label sequence and the best segmentation, $\hat{\mathbf{b}}$, as follows:

$$(\hat{\mathbf{b}}, \hat{\mathbf{h}}) \approx \arg\max_{\mathbf{b},\mathbf{h}} P(\mathbf{h})\, P(\vec{o}_{b_0}^{\,b_1} \mid h_1) \ldots P(\vec{o}_{b_{n-1}}^{\,b_n} \mid h_n) \quad (2)$$

where $\mathbf{b}$ is a sequence of $n+1$ *boundary marks*, $b_0, b_1, \ldots, b_n$, such that $b_0 = 0,\ \ b_i < b_j,\ 1 < i < j < n,\ b_n = L$. These marks define the different lines found in the page. This exactly corresponds to the optimization problem solved by the Viterbi search algorithm [3].

The HMMs (one for each line label) are trained with an instance of the EM algorithm called forward-backward or Baum-Welch re-estimation [3]. It is important to note that to carry out the training, the system only requires the total number of lines in the training pages. Finally, for each detected line, an extraction polygon can be easily calculated by taking some pixels above and below the baseline.

III. Alignment between Segmented Line Images and Line-Level Transcripts

As previously commented, when trying to align a set of detected line images with a set of line-level transcripts, there is no guarantee that the number of lines and transcripts coincide, or that the elements of both sets can be correctly paired sequentially. Transcripts are generally assumed to be sequentially given in the "correct" reading order, but it may be extremely difficult to automatically infer the correct reading order of the detected line image regions. If some "natural" reading order can be assumed in the set of line images, then forced-alignment [9], [5] at page level could be used. Moreover, this would not even need the transcripts to be given at line level (i.e., no synchronized line breaks would be needed).

However, several issues take place during the production of each of the two sets which may lead to disparity. With respect to the line-level transcripts, the main issues are the missing line breaks and the forgetfulness of transcribing small pieces of text such as catch-words, added text, page numbers, etc. Likewise, concerning the line images, there are plenty of special (often small) lines, such as interline added text, that very frequently fail to be correctly detected. Moreover, several kinds of noise typically found in historical documents, such as stains, ink bleed through, etc., are often wrongly detected as line regions. In Figure2 it is shown an example of the disparity that we can found in the two sets. In spite of having the same number of transcribed lines and line images, the pairing is not correct. Whereas in the image the page number has been detected, its transcription has not been included in the text. In addition, the second line detected in the image has been transcribed as two different lines. Therefore, altough the number of lines in the two sets coincide, an automatically pairing of the elements is not possible.

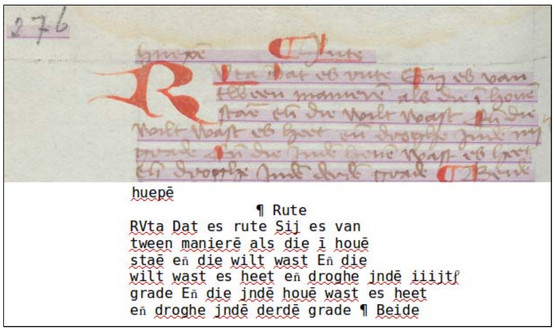

Figure 2. Example of the input to the alignment system.

Before discussing the alignment approach proposed here, two assumptions are taken for granted. First, it is assumed that blocks of both the line images and the line-level transcripts are already correctly paired; and second, it is considered that the elements (lines) within paired blocks are both sorted in some natural reading order. These assumptions are reasonable for the majority of handwritten documents of interest: large volumes with simple page structure for which there are existing transcripts.

The proposed alignment method uses dynamic programming to obtain the best alignment between a given *sequence* of line images and a given *sequence* of line-level transcripts. To this end, we first need to establish a common representation of the data of both sets (i.e. line images and transcripts) which allow pairs of elements of each sequence to be properly compared. For instance, the alignment approach called "morphologic alignment" [14], line images and transcripts are both represented as lengths, the former estimated in terms of number of characters possibly filling the line image region considered and the latter directly by its actual number of characters.

In the alignment method proposed in this work, a line image is represented by the best character string hypotheses obtained by decoding the line image with a HMM-based HTR recognizer. This representation called *HTR hypotheses* from now on, can be straightforwardly compared with a transcript character strings without further changes. Section V-B describes the settings of the HTR recognizer used to obtain HTR hypotheses for the lines images to be aligned.

In more detail, the proposed alignment method is actually a two-level dynamic programming approach. The top level finds an optimal alignment between a sequence of HTR hypotheses and a sequence of transcripts, while the bottom level computes the cost of pairing each particular pair of elements. This cost is given by the well known Levenshtein distance at character level [15]. The top-level best alignment is one which minimizes an objective function defined as the sum of Levenshtein costs of all the matched elements. Since this alignment may contain also insertions and deletions, the costs of these "pseudo-matches" is established as the Levenshtein distance with the empty string of characters; i.e., the number of characters of the either HTR hypothesis or the transcript.

These elementary Levenshtein distance based costs can be easily converted into matching confidence score estimates as follows: $s = 1 - L_d/l_r$, where L_d is the Levenshtein cost between a hypothesis and a line-level transcript, and l_r is the length of the transcript in number of characters.

The optimal top-level alignment, along the corresponding matching confidence scores, provides information as to which matching pairs are actually dependable (and therefore useful for training HTR systems). Moreover, the insertions, deletions and low-confidence matches serve to (hopefully) detect unpaired and/or missing elements of one sequence with respect to the other – an information which allows for cost-effective manual amendment of transcripts and/or line image detection errors.

IV. Handwritten Text Recognition Overview

The HTR system used in this paper follows the classical architecture composed of three main modules: document image preprocessing, line image feature extraction and Hidden Markov Model (HMM) and language model training/decoding [16].

In the preprocessing module, the pages are first divided into line images. Once the lines are detected and extracted appropriate filtering methods are applied to remove noise, improve the quality of the line image and to make the documents more legible [17]. Then, each preprocessed line image is represented as a sequence of feature vectors [18].

Given a handwritten line image, represented by a feature vector sequence, $\mathbf{x}$, the HTR problem can be formulated as the problem of finding a word sequence which most likely corresponds to the text actually written in the image. Using the Bayes' rule this can be writen as:

$$\hat{\mathbf{w}} = \arg\max_{\mathbf{w}} P(\mathbf{w} \mid \mathbf{x}) = \arg\max_{\mathbf{w}} P(\mathbf{x} \mid \mathbf{w})P(\mathbf{w}) \quad (3)$$

$P(\mathbf{x} \mid \mathbf{w})$ is approximated by concatenated character models, usually HMMs [3], while $P(\mathbf{w})$ is approximated by a word language model, usually n-grams [3].

The search (or decoding) to obtain $\hat{\mathbf{w}}$ is optimally carried out using the Viterbi algorithm [3]. See [16] for a more detailed description of this approach to HTR, including text line processing, model training and decoding.

V. Experimental Framework

In this section the corpora used in the experiments, the systems setup and the obtained results are described.

A. Corpus

The experiments have been carried out using the *C5 Hattem Manuscript* [19]. It is a single writer document from the 15th century mainly written in Middle Dutch and composed of 572 leaves.

From these 572 leaves, a ground truth (GT) of 40 pages were manually annotated [19]. That is, the blocks and lines of the pages were manually detected and then the pages were transcribed line by line by a human expert as explained in [19]. These pages are not a consecutive part of the book but instead they were selected from the complete collection by an expert. The selection was performed in order to obtain a reduced set of pages that would be representative of all the different page formats that appear along the book. Figure 3 shows some examples of the selected images.

The 40 pages were divided into 8 blocks of 5 pages each, aimed at performing cross-validation experiments. The average values of the statistics related with the different partitions are shown in column "Ground Truth" of the Table I. The average number of running words for each partition that did not appear in the other seven partitions is shown in the running out-of-vocabulary (Run. OOV) row. The percentage of Run. OOV is 22%.

In addition to these GT pages, there are 263 pages that have been previously transcribed. These transcripts have been carried out at line level, but the graphical information about the position of the lines in the pages is not available. In this paper these pages are going to be used to test the automatic alignment method presented in the previous sections. The transcripts of these pages are composed of 10,093 lines with a vocabulary of around 12k different words. In the last column of Table I the basic statistics of this additional text are shown.

Table I
Basic statistics of the Hattem database of the Hattem dataset.

	Ground Truth		Additional Text
Number of:	Average	Total	Total
Pages	5	40	263
Lines	194	1,552	10,093
Run. words	1,291	10,330	68,474
Run. OOV	282.4	-	-
Lex. OOV	245.5	-	-
Lexicon	588.5	2,751	12,684
Character set size	45	60	83

B. Experimental setup

Experiments were carried out to assess the capabilities of the proposed automatic alignment method to generate training samples for HTR. The partition of the GT pages described in the previous sub-section was adopted for cross-validation experiments. That is, we carried out 8 rounds, with each of the 5-pages partition block used once as test data leaving the remaining seven blocks to be used as training data.

In each round, a text line segmentation system was trained with the seven blocks defined as training data. Then, the additional 263 pages were automatically segmented. The detected lines of each page were automatically transcribed with an HTR system using a special language model in which only the transcripts of the different lines in the page have non null probability (this model is a so called "canonical" finite-state machine with as many parallel branches as lines are in the page transcript). On the other hand, the optical models of this HTR system were trained with the line images of the seven partitions defined as training data. The recognition output was used to carry out an automatic alignment as explained in section III. Those lines whose confidence alignment was above a predefined threshold were included as additional training data (in this work the threshold was set to 1.0). Finally, a new HTR system was trained including the additional training data.

With respect to the text line segmentation, the same feature extraction and HMM meta parameters determined in previous experiments with different handwritten data sets [13] were adopted here. On the other hand, the VLM was fixed on the base of prior knowledge of the general structure of the pages. The train block was used to train the vertical line shape HMMs.

Regarding the HTR system, the line images of the training partition blocks were used to train the corresponding character HMM models using the standard embedded Baum-Welch training algorithm [3]. A left-to-right HMM was trained for each of the elements appearing in the training text images, such as lowercase and uppercase letters, symbols, special abbreviations, possible spacing between words and characters, etc. Each lexical token (word) was modelled by a stochastic finite-state automaton, which

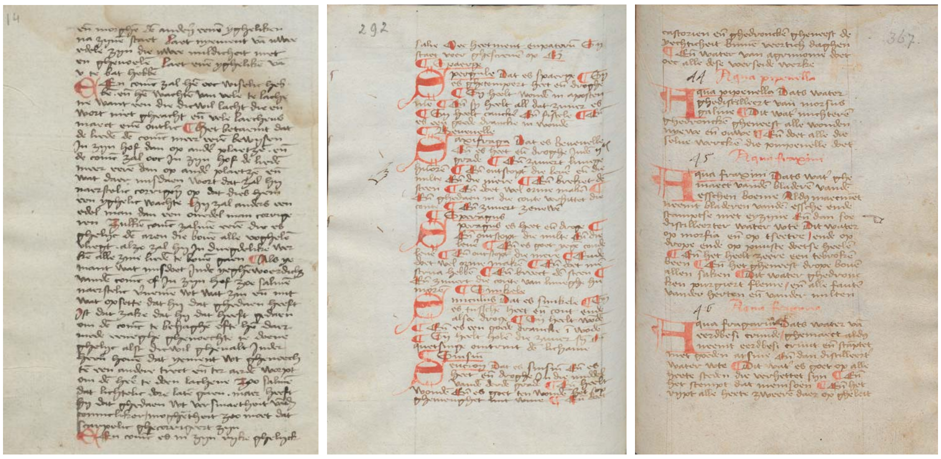

Figure 3. Page image examples of the Hattem collection.

represents all the observed spellings (concatenations of characters) of the word.

Finally, the concatenation of words into line texts was modelled by bi-grams. Two different bi-gram language models with Kneser-Ney back-off smoothing [20] were trained for each round. One trained only with the transcripts of the training data in the seven partitions and another one trained using both the transcripts of the training data and all the text contained in the 263 pages.

C. Assessment Measures

The impact of the automatic alignment in the generation of training data for HTR was assessed in terms of the *Word Error Rate* (WER) and the *Character Error Rate* (CER). They are defined as the minimum number of words/characters that need to be substituted, deleted or inserted to convert the sentences recognized by the system into the reference transcripts, divided by the total number of words/characters in these transcripts.

D. Results

As previously commented, the experiments were aimed at testing whether the automatic alignment method presented in previous sections can be useful to improve the accuracy of an HTR system. To this end, we carried out experiments with different training sets both for optical and for language model training.

Table II shows the results obtained with three different training sets. The first row is considered here has a baseline (B-HTR). It corresponds to an HTR system trained with only the training samples defined in the GT. The second row (LM-HTR) corresponds to an HTR system which used the same optical models that the previous one, but a better language model, straightforwardly trained with all the additional text available (the extra 263 page transcripts). The last row (ALIGN-HTR) corresponds to an HTR system where the optical models were trained using both the GT training set and the automatically aligned lines whose confidence is equal to 1. The language model used in this system is the same used in the second system.

Table II
WER AND CER OBTAINED BY THE DIFFERENT HTR SYSTEM STUDIED.

	WER	CER
B-HTR	33.1	17.6
LM-HTR	26.1	12.2
ALIGN-HTR	24.2	11.1

Taking into account the 8 cross-validation rounds, the number of automatically detected lines in the 263 pages were $9,560$, a bit less than the $10,093$ lines of the corresponding transcripts. This means that about 5.0% of the lines failed to be detected. The detected lines were then aligned with the line transcripts, and the number of perfect alignments (with confidence equal to 1.0) which were included in the training set was around $3,000$ for all partitions. This means that about 30% of the additional line images available were actually used for training in the ALIGN-HTR experiment.

As expected, using only the pure textual information of the extra transcripts to straightforwardly train a better language model, improved the HTR accuracy significantly. This improvement was mainly due to the reduction of the running OOV words that in B-HTR experiment was about 22.0%, and in the LM-HTR experiment was about 8.0%. On the other hand, when the automatically aligned lines were included in the training set for improving the optical models, an additional improvement of two points was obtained (ALIGN-HTR). This more modest improvement may suggests that most of the added training pairs were correctly aligned.

To explain the fact that using as much as $3,000$ additional training pairs only an improvement of 2.0 points in WER was obtained, we should take into account that all the process has been automatically carried out, without human supervision. Therefore, it can occur that some of the lines included in the training set were not actually aligned correctly. These samples degenerate the HMMs instead to improve them. In Figure 4 we can see some examples of correct and erroneous alignments. In any case, from the results it is shown that using the automatic alignment can impact positively in the HTR results, without involving extra human effort.

Figure 4. Alignment examples. The first one is a correct alignment. The second one has been correctly aligned but the line is not perfectly detected, there is a big blank space at the end. The last one is an incorrect alignment of an incorrectly segmented line.

VI. Conclusions and future work

The generation of training samples is one of the major bottlenecks in the development of accurate HTR system. Even when the transcript of a handwritten document is available, putting in correspondence the physical lines in the images with the lines in the transcripts can be problematic. In this paper we have presented a method for automatically aligning handwritten text images and their respective transcripts.

Experiments have been carried out in a medieval manuscript where, in addition of a perfectly annotated GT, a relatively large quantity of pages which were transcribed, but not aligned, were available. From the results we can conlude that using the proposed automatic alignment method, significant improvements can be achieved. The WER has been improved 2 point with respect to using only the transcriptions of the additional pages to generate a new language model. Finally, it is important to remark here that the automatic alignment do not involve any extra human effort, given that all the process is carried out automatically without human supervision.

In a future work we plan to study how the confidence of the alignment affect in the results. In the experiments presented in this paper, only "perfect" alignments, with confidence 1.0, were considered. However, we plan to evaluate how much information can be obtained using a larger number of lower confidence alignments.

Acknowledgment

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